

Dirichlet-Multinomial Error-Choice Simulation ($p_{\min} = 0.01$)

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1 Overview

This report evaluates **dirichlet-multinomial** sampling error across cell-type proportions generated from a weighted $\text{Beta}(\alpha, 1)$ distribution, **with the smallest population proportion set at $p_{\min} = 0.01$** .

1.1 Success rule

For each replicate and cell type j , the estimated proportions are averaged over the N people in that replicate. The average is compared with the population-level true proportion p_j :

$$\bar{p}_j = \frac{1}{N} \sum_{i=1}^N \hat{p}_{ij}, \quad AE_j = |\bar{p}_j - p_j|, \quad ARE_j = \frac{|\bar{p}_j - p_j|}{p_j}. \quad (1)$$

A replicate **succeeds** for AE if $\max AE_j \leq \tau_{AE}$, and for ARE if $\max ARE_j \leq \tau_{ARE}$. AE and ARE are always evaluated separately and never combined into one criterion.

2 Simulation setup

The data are generated in two steps (a Dirichlet-multinomial hierarchy):

1. Person i has a true composition $\theta_i \sim \text{Dirichlet}(c \cdot p)$, where p is the population mean composition and c is the concentration.
2. From that person, n cells are sampled: $\text{counts}_i \sim \text{Multinomial}(n, \theta_i)$, so $\hat{p}_{ij} = \text{count}_{ij}/n$.

The population proportions p are fixed values derived from a Beta distribution, with the smallest proportion pinned (Table 2, Figure 1). The construction starts from the plain $\text{Beta}(\alpha, 1)$ proportions on a grid of K points. The smallest type, and every type that would fall below $p_{\min}$, is set to exactly $p_{\min}$. The remaining types are rescaled proportionally so that all proportions still sum to one. A larger α makes the composition more skewed, so more types are set to $p_{\min}$: 1, 2, 3, 4 types for $\alpha = 2, 3, 4, 5$ respectively. Each scenario (α, N, n) is simulated B times. Table 1 lists all settings.

Table 1: Simulation settings (`simulation_dm_errorchoice_pmin_defaults()`).

Parameter	Value
alpha (Beta shape)	2, 3, 4, 5
K (cell types)	10
B (replicates per scenario)	1,000
Concentration c	10000
Proportion method	fixed_min_beta
p_min	0.01
N (people per replicate)	1, 2, 3, 5, 10
n per person, sweep	100 to 100,000,000 (25 log-spaced values)
n per person, threshold sweep	200,000
tau_AE (fixed)	0.01
tau_ARE (fixed)	0.5
Target success rate	0.95
Seed	260926

Table 2: Population mean proportions p_j per cell type (columns) for each α (rows). The smallest type and every type that would fall below $p_{\min} = 0.01$ are set to exactly 0.01, so 1, 2, 3, 4 types are set to $p_{\min}$ for $\alpha = 2, 3, 4, 5$ respectively.

alpha	1	2	3	4	5	6	7	8	9	10
2	0.01	0.03	0.05	0.07	0.09	0.11	0.13	0.15	0.17	0.19
3	0.01	0.01	0.019	0.036	0.06	0.09	0.13	0.17	0.21	0.27
4	0.01	0.01	0.01	0.017	0.036	0.065	0.11	0.17	0.24	0.34
5	0.01	0.01	0.01	0.01	0.02	0.045	0.087	0.15	0.26	0.4

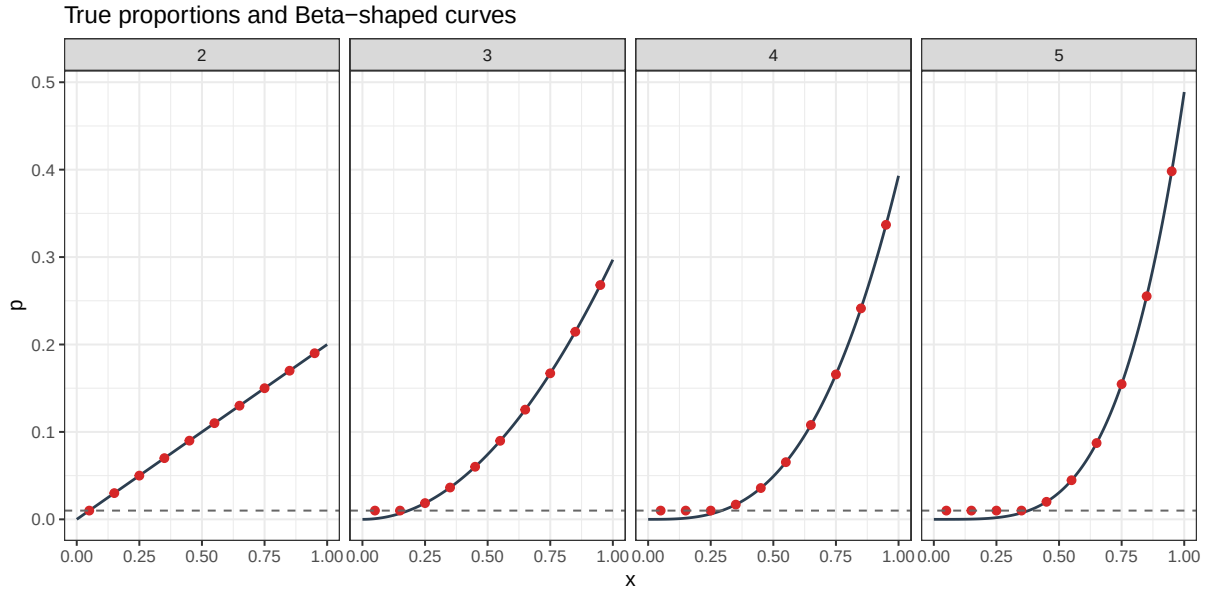


Figure 1: Construction of the population proportions: points are the proportions p_j per cell type, the curve is the underlying scaled Beta curve, and the dashed line marks $p_{\min} = 0.01$. One panel per α .

3 Simulation results

3.1 Success rate versus threshold

Figure 2 and Figure 3 show the success rate as a function of the threshold τ at a fixed 200,000 cells per person.

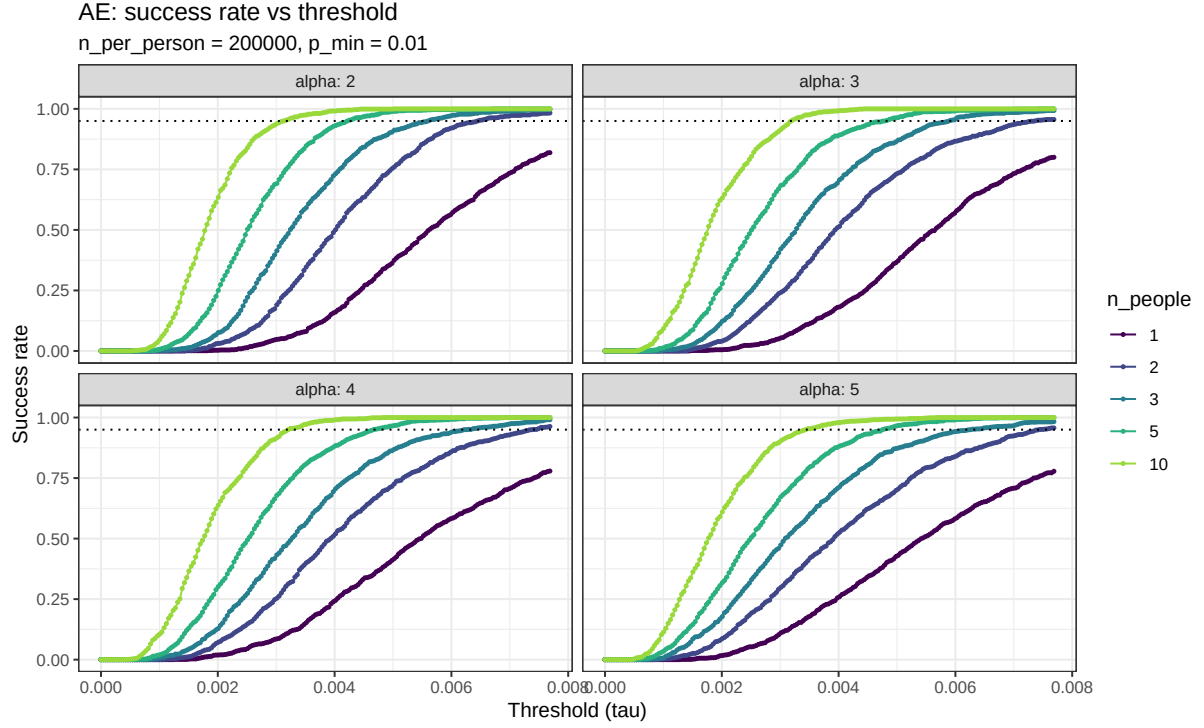


Figure 2: AE success rate versus threshold τ_{AE} at 200,000 cells per person. One panel per α , one colour per number of people N ; the dotted line marks the 95% target.

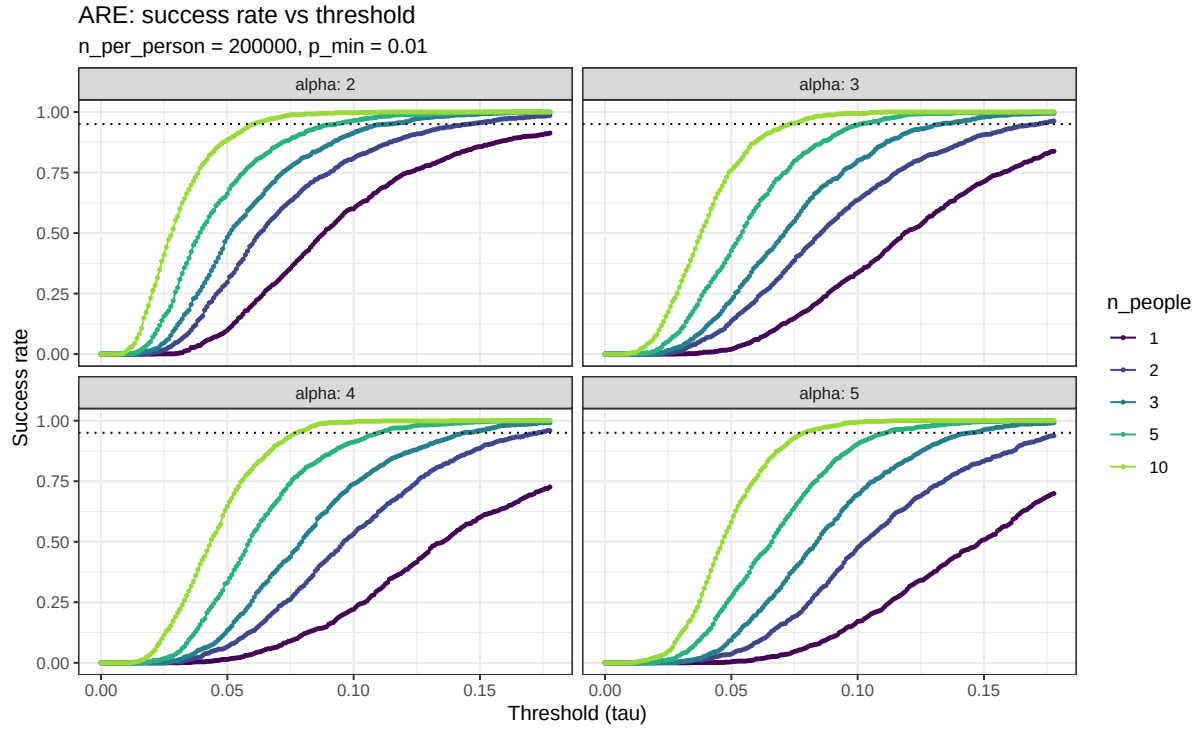


Figure 3: ARE success rate versus threshold τ_{ARE} at 200,000 cells per person.

3.2 Success rate versus cells per person

Figure 4 and Figure 5 fix the threshold ($\tau_{AE} = 0.01$, $\tau_{ARE} = 0.5$) and vary the number of cells per person.

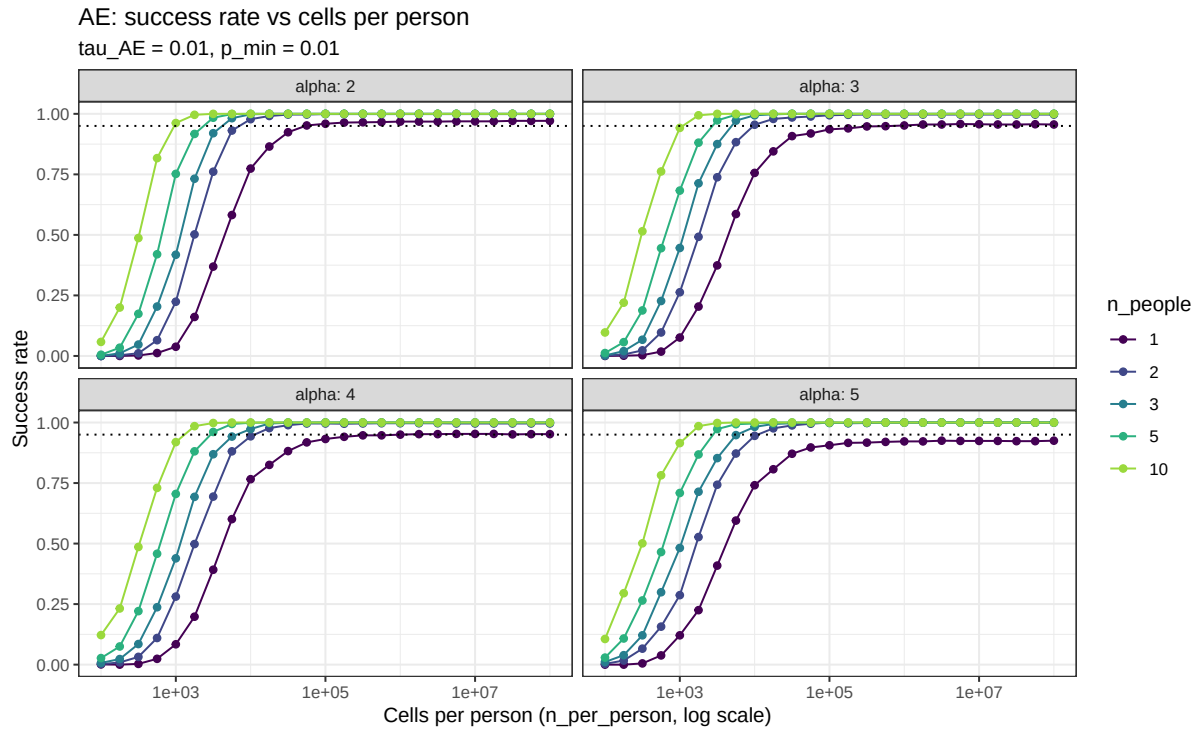


Figure 4: AE success rate versus cells per person (log scale) at $\tau_{AE} = 0.01$. Success rises with cells per person and then reaches a plateau.

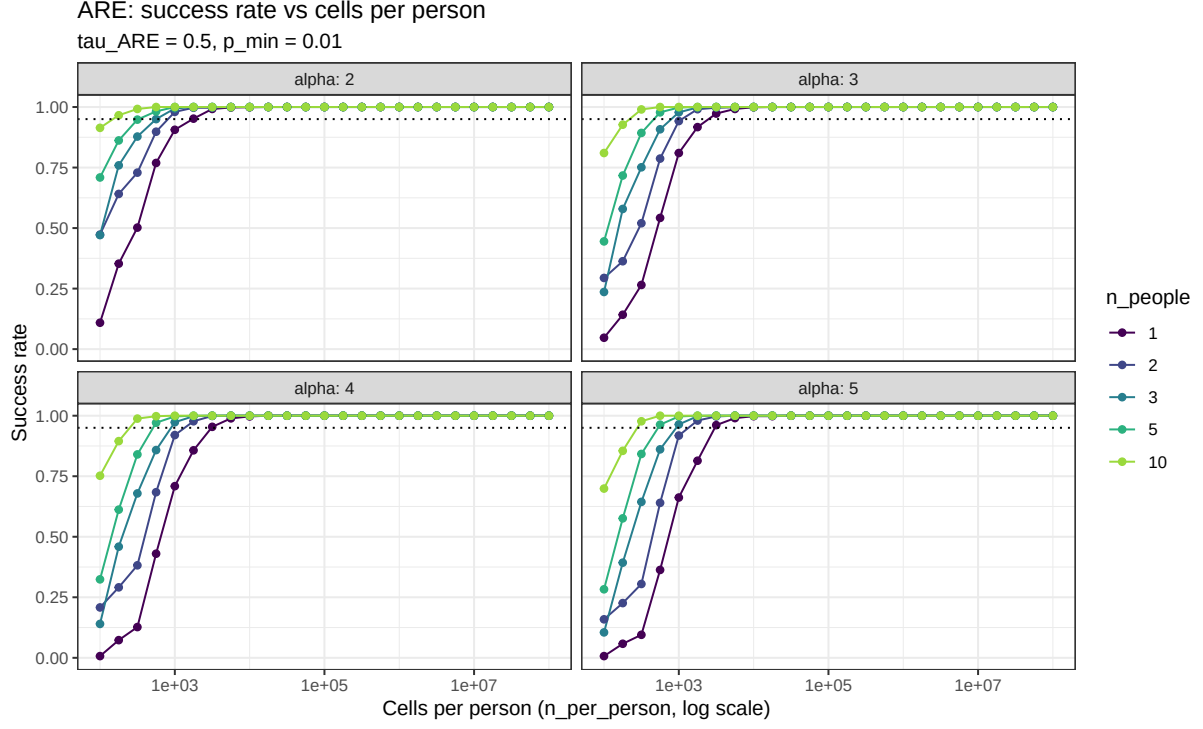


Figure 5: ARE success rate versus cells per person (log scale) at $\tau_{ARE} = 0.5$.

Compare Figure 4 and Figure 5 with their corresponding figures in the previous report `simulation_dm_errorchoice.pdf`. The curves for AE look similar, but the curves for ARE are clearly impacted. This is as expected, as for the ARE the smallest proportions are more impactful.

3.3 Required cells per person

Figure 6 and Figure 7 show, for each α and number of people N , the smallest number of cells per person for which the success rate reaches the 95% target at the fixed thresholds, found with an iterative sample-size solver. Before solving, it is checked whether the target can be reached at all with very many cells per person; combinations where it cannot are marked with $\times$ at the top of the panel. Open markers indicate that the sample size calculation algorithm did not converge in the allotted time. The marker indicates the last value.

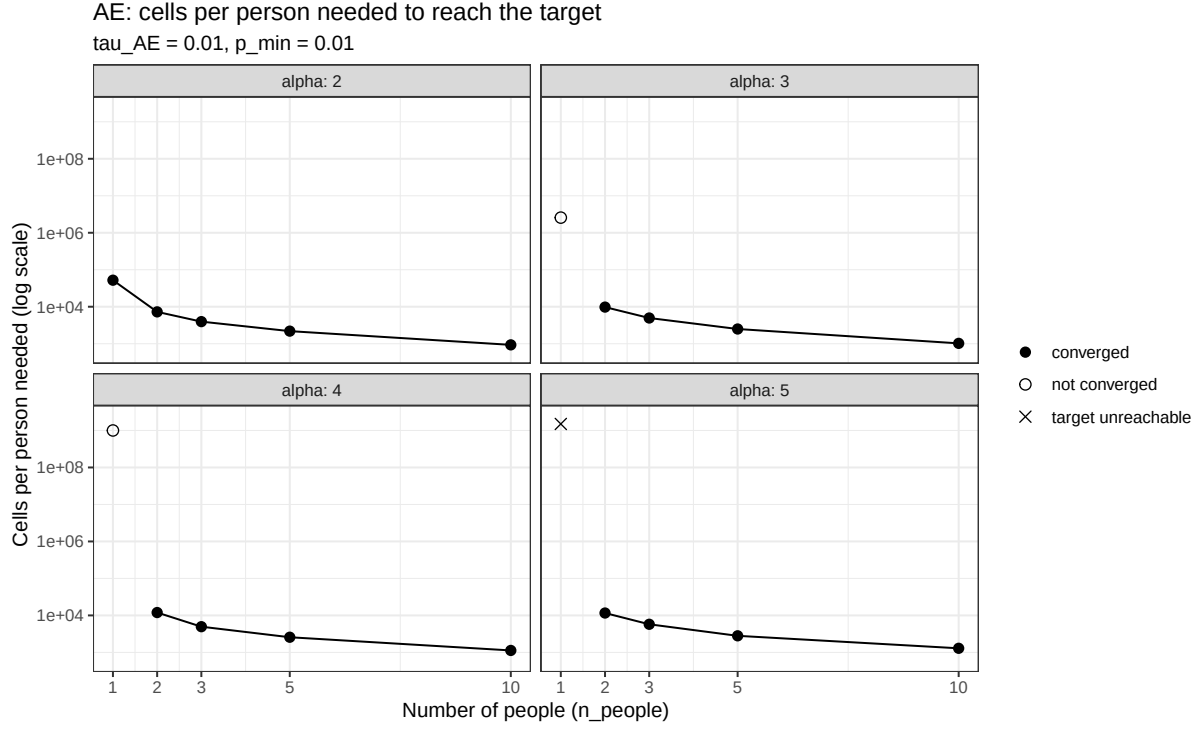


Figure 6: Cells per person needed to reach the 95% AE success target at $\tau_{AE} = 0.01$ (log scale), versus the number of people N . One panel per α . An open marker means the solver did not converge; $\times$ at the top means the target cannot be reached.

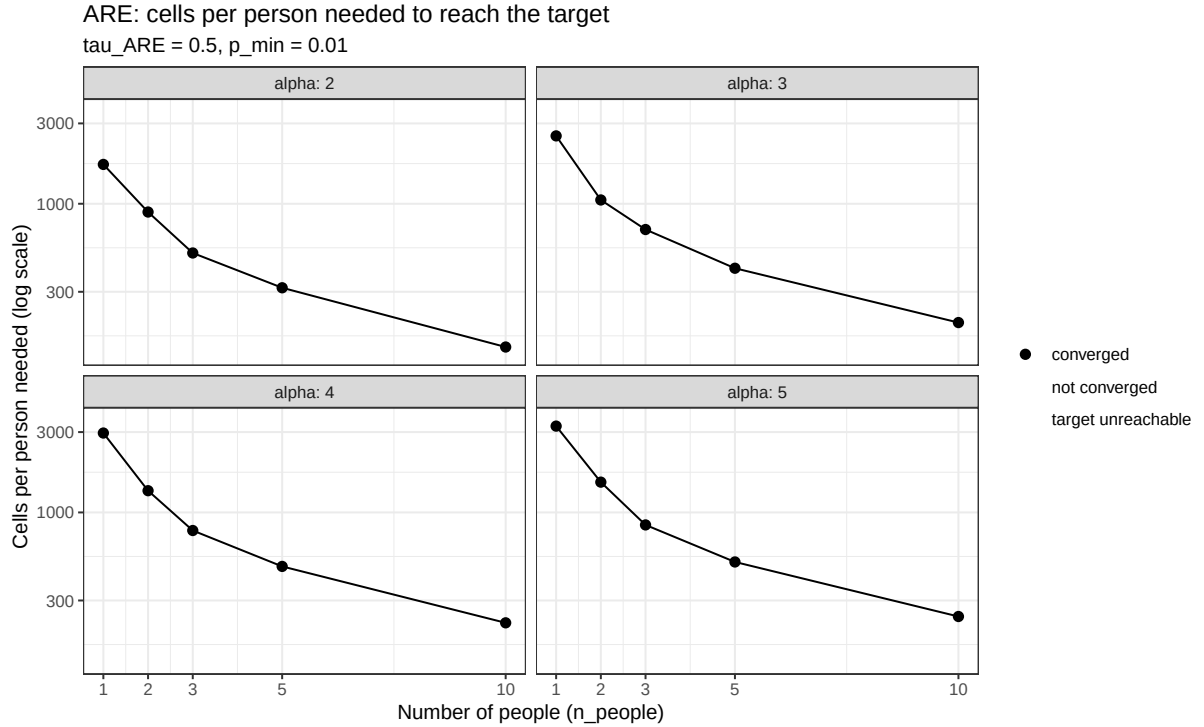


Figure 7: Cells per person needed to reach the 95% ARE success target at $\tau_{ARE} = 0.5$ (log scale), versus the number of people N . One panel per α . An open marker means the solver did not converge; $\times$ at the top means the target cannot be reached.